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Plasma processing apparatus

Abstract

There is a plasma processing apparatus, comprising: a plasma processing chamber; a substrate support having a ring supporting surface; an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the through holes having upper and lower hole portions, the lower hole portion having a flaring shape; a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the grooves corresponding to the through holes; at least three lift pins disposed below the ring supporting surface, the lift pins corresponding to the grooves, each of the lift pins having upper and lower supporting portions, the upper supporting portion being configured to support the conductive ring, the lower supporting portion configured to support the insulating ring; and at least one actuator configured to vertically move the pins.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application No. 2021-215091 filed on Dec. 28, 2021 and Japanese Patent Application No. 2022-163154 filed on Oct. 11, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a plasma processing apparatus.

BACKGROUND

(3) Japanese Laid-open Patent Publication No. 2021-141313 discloses a substrate support having an annular member mounting surface on which an annular member is mounted, and three or more lifters which are configured to protrude from the annular member mounting surface. According to the description in the Japanese Laid-open Patent Publication, a concave portion formed by a concave surface which is concave upward is provided at a location on a bottom of the annular member corresponding to each lifter, and when viewed from a plane, the concave portion is larger than a transfer precision of the annular member upward of the annular member mounting surface,

and is formed to be larger than an upper end portion of the lifter. Further, the upper end portion of the lifter is formed in a hemispherical shape, and the concave surface has a smaller curvature than the concave surface forming the hemispherical shape of the upper end portion of the lifter.

SUMMARY

(4) The technique of the present disclosure provides a plasma processing apparatus that can properly position a ring assembly to a substrate support.

(5) In accordance with an aspect of the present disclosure, there is a plasma processing apparatus, comprising: a plasma processing chamber; a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface; an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a radial direction of the insulating ring and a second width in a circumferential direction of the insulating ring, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side; a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a radial direction of the conductive ring and a fourth width in a circumferential direction of the conductive ring, the fourth width being smaller than the third width; at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting portion being configured to support the conductive ring from downward by contacting a bottom of the groove of the conductive ring through the through hole of the insulating ring, the lower supporting portion configured to support the insulating ring from downward by contacting an inclined surface of the insulating ring, the inclined surface defining the lower hole portion; and at least one actuator configured to vertically move the at least three lift pins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The objects and features of the present disclosure will become apparent from the following description of embodiments, given in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is an explanatory diagram illustrating an outline of a configuration of a plasma processing system;

(3) FIG. 2 is a longitudinal cross-sectional view illustrating an outline of a configuration of a plasma processing apparatus according to an exemplary embodiment;

(4) FIG. 3 is a partial enlarged view of FIG. 2;

(5) FIG. 4A is an explanatory diagram illustrating an outline of a configuration of an insulating ring;

(6) FIG. 4B is an explanatory diagram illustrating the outline of the configuration of the insulating ring;

(7) FIG. 5A is an explanatory diagram illustrating the outline of the configuration of an inner edge ring;

(8) FIG. 5B is an explanatory diagram illustrating the outline of the configuration of the inner edge ring;

(9) FIG. 6A is an explanatory diagram illustrating another formation example of a groove for the inner edge ring;

(10) FIG. 6B is an explanatory diagram illustrating another formation example of the groove for

the inner edge ring;

(11) FIG. 6C is an explanatory diagram illustrating another formation example of a through hole for the insulating ring;

(12) FIG. 7 is a diagram illustrating a form of replacement processing of a ring assembly;

(13) FIG. 8 is a diagram illustrating the form of the replacement processing of the ring assembly;

(14) FIG. 9 is a diagram illustrating the form of the replacement processing of the ring assembly;

(15) FIG. 10A is a diagram illustrating the form of the replacement processing of the ring assembly;

(16) FIG. 10B is a diagram illustrating the form of the replacement processing of the ring assembly;

(17) FIG. 11A is a diagram illustrating another configuration example of the inner edge ring;

(18) FIG. 11B is a diagram illustrating another configuration example of the inner edge ring;

(19) FIG. 11C is a diagram illustrating another configuration example of the inner edge ring;

(20) FIG. 12A is an explanatory diagram illustrating another formation example for the groove for the inner edge ring;

(21) FIG. 12B is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

(22) FIG. 12C is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

(23) FIG. 12D is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

(24) FIG. 13A is an explanatory diagram illustrating another formation example for the groove for the inner edge ring;

(25) FIG. 13B is an explanatory diagram illustrating another formation example of the groove for the inner edge ring; and

(26) FIG. 13C is an explanatory diagram illustrating another formation example of the groove for the inner edge ring.

DETAILED DESCRIPTION

(27) In a manufacturing process of semiconductor devices, plasma processing such as etching or the like is performed on a semiconductor substrate (hereinafter, simply referred to as “substrate”) using plasma. The plasma processing is performed in a state in which a substrate is mounted on a substrate support placed in a processing container configured to be decompressed.

(28) In addition, the substrate support includes a plurality of annular members placed to surround around the substrate on the substrate support in order to obtain an excellent and uniform processing result at a center and a periphery of the substrate at the time of the plasma processing. The plurality of annular members includes an edge ring placed adjacent to the substrate on the substrate support or a cover ring placed to cover an outer surface of the edge ring. Since the annular members are consumed by being exposed to plasma, the annular members need to be regularly replaced. The replacement of the annular member is executed, for example, by using a lifter which is lifted while supporting the annular member and a transfer mechanism transferring the annular member.

(29) Here, when replacing the annular members using the lifter or transfer mechanism, the annular members may not be appropriately placed at desired positions on the substrate support due to transfer precision by the transfer mechanism, etc., for example. In addition, in recently implementing a transfer function of the annular member, enhancement of installation precision of the annular member on the substrate support is further required than the transfer precision by the transfer mechanism.

(30) In the plasma processing apparatus disclosed in the Japanese Laid-open Patent Publication, a groove is formed on a bottom of an edge ring, the groove being larger than the transfer precision by the transfer mechanism, and a through hole is formed in the cover ring, the through hole being larger than the transfer precision by the transfer mechanism, thereby enhancing positioning

precision of the annular member.

(31) However, when the replacement of the annular member is executed by using the plasma processing apparatus described in the Japanese Laid-open Patent Publication, for example, there is a risk that the positioning precision of the annular member will deteriorate or damage the lifter or the annular member when a relative dislocation occurs between positions of the lifter and the groove (through hole) due to a mechanical difference (e.g., processing tolerance) or thermal expansion of the annular member.

(32) The technique according to the present disclosure is derived by considering the circumstances, and provides a plasma processing apparatus that can properly position a ring assembly to the substrate support. Hereinafter, a configuration of a plasma processing system including a substrate processing apparatus according to the exemplary embodiment will be described with reference to the drawings. Further, in the present specification and drawings, the same reference numerals are given to elements having substantially the same functional configuration, so a redundant description will be omitted.

(33) <Plasma Processing System>

(34) In an exemplary embodiment, a plasma processing system includes a plasma processing apparatus **1**, a transfer apparatus **2**, and a controller **3**, as illustrated in FIG. **1**. The plasma processing system is an example of a substrate processing system, and the plasma processing apparatus **1** is an example of a substrate processing apparatus. The plasma processing apparatus **1** includes a plasma processing chamber **10**, a substrate support **11**, and a plasma generator **12**. The plasma processing chamber **10** has a plasma processing space. Further, the plasma processing chamber **10** includes at least one gas inlet for supplying at least one processing gas to the plasma processing space and at least one gas outlet for exhausting gases from the plasma processing space. The gas inlet is connected to a gas supply **20** to be described below and the gas outlet is connected to an exhaust system **40** to be described below. The substrate support **11** is disposed in the plasma processing space, and has a substrate supporting surface for supporting a substrate.

(35) The plasma generator **12** is configured to generate a plasma from at least one processing gas supplied in the plasma processing space. The plasma formed in the plasma processing space may be a capacitively coupled plasma (CCP), an inductively coupled plasma (ICP), an electron-cyclotron-resonance plasma (ECR plasma), a helicon wave plasma (HWP), or a surface plasma (SWP). Further, various types of plasma generator may be used, which include an alternating current (AC) plasma generator and a direct current (DC) plasma generator. In an exemplary embodiment, an AC signal (AC power) used in the AC plasma generator has a frequency in a range of 100 kHz to 10 GHz. Therefore, the AC signal includes a radio frequency (RF) signal and a microwave signal. In an exemplary embodiment, the RF signal has a frequency in a range of 100 kHz to 150 MHz.

(36) The transfer apparatus **2** includes a transfer arm **2a** that holds and transfers a substrate **W** and a ring assembly **120** to be described below. A wafer is an example of the substrate **W**. The transfer apparatus **2** is configured to transfer the substrate **W** and the ring assembly **120** between the outside of the plasma processing apparatus **1** and the substrate support **11** placed inside the plasma processing apparatus **1**.

(37) The controller **3** processes a computer executable command which allows the plasma processing apparatus **1** and the transfer apparatus **2** to execute various processes described in the present disclosure. The controller **3** may be configured to control each element of the plasma processing apparatus **1** and the transfer apparatus so as to execute various processes described herein. In an exemplary embodiment, a part or the entirety of the controller **3** may be included in the plasma processing apparatus **1**. The controller **3** may include a processor **3a1**, a storage **3a2**, and a communication interface **3a3**. The controller **3** may be implemented by a computer **3a**, for example. The processor **3a1** may be configured to perform various control operations by reading a program from the storage **3a2** and executing the read program. The program may be stored in the

storage **3a2** in advance or obtained through a medium as necessary. The obtained program is stored in the storage **3a2**, and read and executed from the storage **3a2** by the processor **3a1**. The medium may be various storage media readable by the computer **3a** or a communication line connected to the communication interface **3a3**. The processor **3a1** may be a central processing unit (CPU). The storage **3a2** may include a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or a combination thereof. The communication interface **3a3** may communicate with the plasma processing apparatus **1** through a communication line such as local area network (LAN), etc. Further, the storage media may be transitory or non-transitory.

(38) <Plasma Processing Apparatus>

(39) Next, as an example of the plasma processing apparatus **1**, an example of a configuration of a capacitively coupled plasma processing apparatus **1** will be described. FIG. **2** is a longitudinal cross-sectional view illustrating an outline of a configuration of a plasma processing apparatus **1** according to an exemplary embodiment. FIG. **3** is a partial enlarged view enlarging and illustrating a part of a configuration of a substrate support **11** illustrated in FIG. **2**.

(40) The capacitively coupled plasma processing apparatus **1** includes a plasma processing chamber **10**, a gas supply **20**, a power supply **30**, and an exhaust system **40**. Further, the plasma processing apparatus **1** includes the substrate support **11** and a gas introduction unit. The gas introduction unit is configured to introduce at least one processing gas into the plasma processing chamber **10**. The gas introduction unit includes a showerhead **13**. The substrate support **11** is placed in the plasma processing chamber **10**. The showerhead **13** is placed above the substrate support **11**. In an exemplary embodiment, the showerhead **13** constitutes at least a part of a ceiling of the plasma processing chamber **10**. The plasma processing chamber **10** has a plasma processing space **10s** defined by the showerhead **13**, a side wall **10a** of the plasma processing chamber **10**, and the substrate support **11**. The plasma processing chamber **10** is grounded. The showerhead **13** and the substrate support **11** are electrically insulated from a housing of the plasma processing chamber **10**.

(41) The substrate support **11** includes a body **110**, a ring assembly **120**, and a lifter **130**. The body **110** has a central region **110a** for supporting the substrate **W** and an annular region **110b** for supporting the ring assembly **120**. The annular region **110b** of the body **110** surrounds the central region **110a** of the body **110** in a plan view. The substrate **W** is disposed on the central region **110a** of the body **110**, and the ring assembly **120** is placed on the annular region **110b** of the body **110** to surround the substrate **W** on the central region **110a** of the body **110**. Therefore, the central region **110a** is also called a substrate supporting surface for supporting the substrate **W**, and the annular region **110b** is also called a ring supporting surface for supporting the ring assembly **110**.

(42) As illustrated in FIG. **3**, in an exemplary embodiment, the body **110** includes a base **111**, an electrostatic chuck **112**, a support **113**, and an insulator **114**.

(43) The base **111** includes a conductive member. The conductive member of the base **111** may serve as a lower electrode. The electrostatic chuck **112** is placed on the base **111**. The electrostatic chuck **112** includes a ceramic member **112a** and an electrostatic electrode **112b** placed in the ceramic member **112a**. The ceramic member **112a** has the central region **110a**. In an exemplary embodiment, the ceramic member **112a** and the support **113** have the annular region **110b**. Further, another member surrounding the electrostatic chuck **112** may have the annular region **110b** such as an annular electrostatic chuck instead of an annular insulating member such as the support **113**. The ring assembly **120** may be placed on the annular electrostatic chuck or the annular insulating member or placed on both the electrostatic chuck **112** and the annular insulating member. Alternatively, the constitution of the annular electrostatic chuck or the annular insulating member is omitted, and the ring assembly **120** may be placed only on the electrostatic chuck **112**.

(44) Further, at least one RF/DC electrode coupled to a radio frequency (RF) power supply **31** and/or direct current (DC) power supply **32** to be described below may be placed in the ceramic member **112a**. In this case, at least one RF/DC electrode serves as the lower electrode. When a bias

RF signal and/or DC signal to be described below is supplied to at least one RF/DC electrode, the RF/DC electrode is also called a bias electrode. Further, the conductive member of the base **111** and at least one RF/DC electrode may serve as a plurality of lower electrodes. Further, the electrostatic electrode **112b** may serve as the lower electrode. Therefore, the substrate support **111** includes at least one lower electrode.

(45) The support **113** is a member formed in a ring shape when viewed on the plan view by using an insulating material such as quartz, etc., for example and supports the base **111** and the electrostatic chuck **112**. In an exemplary embodiment, the ring assembly **120** is mounted on the support **113** as described above. Further, a plurality of, in the exemplary embodiment, three through holes **113a** penetrating in a thickness direction are formed in the support **113**. A lift pin **131** to be described below of the lifter **130** is inserted into the through hole **113a** as illustrated in FIG. 3.

(46) The insulator **114** is a cylindrical member made of ceramic, etc., and supports the support **113**. The insulator **114** is formed to have an outer diameter equal to an outer diameter of the support **113**, for example, and supports a periphery of the support **113**.

(47) The ring assembly **120** as the annular member includes a plurality of annular members. As illustrated as an example in FIG. 3, the ring assembly **120** includes an insulating ring **121**, an outer edge ring **122**, and an inner edge ring **123**, as the plurality of annular members. The outer edge ring **122** and the inner edge ring **123** are made of silicon, silicon carbide, or quartz, as an example. The insulating ring **121** is supported on the support **113** constituting the annular region **110b** (ring supporting surface). The inner edge ring **123** is supported on the electrostatic chuck **112** and the insulating ring **121** constituting the annular region **110b** (ring supporting surface). That is, the insulating ring **121** and the inner edge ring **123** are supported on the support **113** constituting the annular region **110b** (ring supporting surface) in order. Further, the ring assembly **120** is lifted up from the annular region **110b** (ring supporting surface) by an operation of the lifter **130**, and thus configured to be transferred by the transfer apparatus **2** placed outside the plasma processing apparatus **1**.

(48) Further, the inner edge ring **123** may constitute a “conductive ring” according to the technique of the present disclosure. Further, the outer edge ring **122** may constitute an “additional conductive ring” according to the technique of the present disclosure. In other words, the ring assembly **120** as the annular member includes an insulating ring **121**, an additional conductive ring **122**, and a conductive ring **123** as a plurality of annular members.

(49) The outer edge ring **122** is formed large in the thickness direction as compared with the insulating ring **121**, and placed to surround the inner edge ring **123** and the insulating ring **121**, and placed so that an inner part of the outer edge ring **122** overlaps with an outer part of the insulating ring **121** in a vertical direction. In other words, an upper surface of the insulating ring **121** constitutes a support surface for the outer edge ring **122** and the inner edge ring **123**.

(50) Further, the configuration of the ring assembly **120** is not limited to an illustrated example, and for example, the insulating ring **121** and the outer edge ring **122** may be integrally configured. In this case, the insulating ring **121** and the outer edge ring **122** may be configured by the same member, that is, the outer edge ring **122** may serve as a part of the insulating ring and the insulating ring may serve as a part of the additional conductive ring.

(51) Further, a plurality of, in the exemplary embodiment, three through holes **121a** penetrating in the thickness direction are formed in the insulating ring **121** as illustrated in FIGS. 4A and 4B. The through hole **121** is formed at a position corresponding to the through hole **113a** formed in the support **113**, and an upper supporting portion **131a** of the lift pin **131** to be described below is inserted into the through hole **121a**.

(52) Three through hole **121a** has an upper hole portion **121a1** and a lower hole portion **121a2**. The upper hole portion **121a1** of the through hole **121a** has a substantially rectangular shape in which a first width **W1** which extends in a radial direction **R1** of the insulating ring **121** is formed to be larger than a second width **W2** which extends in a circumferential direction **C1** of the insulating

ring **121** orthogonal to the radial direction **R1**. The first width **W1** is formed to be larger than at least a diameter (a fifth width **W5** to be described below) of the upper supporting portion **131a** of the lift pin **131** to be described below. The second width **W2** is formed to be larger than the diameter (fifth width **W5**) of the upper supporting portion **131a**, and further, formed to be smaller than a diameter (sixth width **W6** to be described below) of a lower supporting portion **131b** of the lift pin **131** to be described below.

(53) The lower hole portion **121a2** of the insulating ring **121** has a flare shape (inclined surface **1211a**) in which the first width **W1** and the second width **W2** of the upper hole portion **121a1** spread downward (toward a bottom of the insulating ring **121**). Preferably, a lower end of the inclined surface **1211a** of the lower hole portion **121a2** is formed to be at least larger than the transfer precision of the ring assembly **120** by the transfer apparatus **2**.

(54) In the insulating ring **121** according to the exemplary embodiment, therefore, the upper supporting portion **131a** of the lift pin **131** is inserted into the upper hole portion **121a1**, and the lower supporting portion **131b** of the lift pin **131** comes into contact with two points on the inclined surface **1211a** of the lower hole portion **121a2** in the circumferential direction **C1**.

(55) Further, as described above, in the upper hole portion **121a1**, the first width **W1** which extends in the radial direction **R1** is formed to be larger than the second width **W2** which extends in the circumferential direction **C1**, but the sizes of the first width **W1** and the second width **W2** are preferably set to the same as each other as possible. In other words, the upper hole portion **121a1** preferably has a substantially square shape when viewed in the plan view.

(56) The inner edge ring **123** as the conductive ring may also be called a focus ring, and enhances in-plane uniformity of the plasma processing for the substrate **W**. Further, the outer edge ring **122** as the additional conductive ring substantially expands an area of a conductive part (inner edge ring **123**) viewed in the plane on the substrate support **11**. The inner edge ring **123** and the outer edge ring **122** may be made of the silicon, the silicon carbide, or the quartz, as an example.

(57) A plurality of, in the exemplary embodiment, three grooves **123a** are formed on the bottom of the inner edge ring **123** as illustrated in FIGS. 5A and 5B. The groove **123a** is formed at a position corresponding to the through hole **121a** formed in the insulating ring **121**, and a tip of the upper supporting portion **131a** of the lift pin **131** inserted into the through hole **121a** contacts the bottom **123a1**.

(58) Three grooves **123a** have a substantially long hole shape in which a third width **W3** which extends in a radial direction **R2** of the inner edge ring **123** is formed to be larger than a fourth width **W4** which extends in a circumferential direction **C2** of the inner edge ring **123** orthogonal to the radial direction **R2**. The third width **W3** is formed to be larger than a diameter (a fifth width **W5**) of the upper supporting portion **131a**. The fourth width **W4** is formed to have the substantially same size as the diameter (the fifth width **W5** to be described below) of the upper supporting portion **131a** of the lift pin **131** to be described below.

(59) Further, the bottom **123a1** of three grooves **123a** has a shape of being fitted into a tip shape of the upper supporting portion **131a** of the lift pin **131**, in particular, in an extension direction of the fourth width **W4**. Specifically, in the exemplary embodiment, as described below, the upper supporting portion **131a** has a hemispherical shape **131a1** in which the tip protrudes as described below, but the bottom **123a1** of the groove **123a** is formed in a concave shape (hereinafter, the bottom **123a1** having a curve shape may be referred to as a “curved bottom **123a1**”) having the substantially same curvature as the hemispherical shape **131a1**.

(60) An inclined portion is formed on a flare-shaped side surface **123a2** of three grooves **123a** so that the third width **W3** and the fourth width **W4** spread downward (toward the bottom of the inner edge ring **123**). A lower end of the inclined surface **1211a** of the groove **123a** is preferably formed to be at least larger than the transfer precision of the inner edge ring **123** by the transfer apparatus **2**.

(61) In the inner edge ring **123** according to the exemplary embodiment, therefore, the upper

supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** in a circumferential direction (a fourth width (**W4**) direction) of the inner edge ring **123**.

(62) Further, a depth of the groove **123a** is 0.4 mm or more and 1.0 mm or less, as an example.

(63) In an exemplary embodiment, the lifter **130** includes a plurality of, in the exemplary embodiment, three lift pins **131** corresponding to the through holes **121a** of the insulating ring **121** and the grooves **123a** of the inner edge ring **123**, and at least one actuator **132**. Further, the lift pin **131** includes a plurality of cylinder supporting portions having different diameters. As illustrated as an example in FIG. 3, the lift pin **131** includes the upper supporting portion **131a** and the lower supporting portion **131b**. The upper supporting portion **131a** and the lower supporting portion **131b** may be integrally configured.

(64) The upper supporting portion **131a** at least has the fifth width **W5** having a diameter smaller than the first width **W1** and the second width **W2** of the through hole **121a** formed in the insulating ring **121**. The fifth width **W5** has the substantially same size as the fourth width **W4** of the groove **123a** formed in the inner edge ring **123**. The upper supporting portion **131a** is concatenated in an axial direction from the upper surface of the lower supporting portion **131b** to be described below, and moves in a vertical direction (axial direction) integrally with the lower supporting portion **131b** by the operation of the actuator **132**. In addition, the upper supporting portion **131a** is configured to be projected and depressed from the upper surface of the insulating ring **121** via the through hole **121a**, and therefore, supports the bottom of the inner edge ring **123** supported on the upper surface of the insulating ring **121**, more specifically, the bottom **123a1** of the groove **123a** formed in the inner edge ring **123**, and moves (lifts up) the bottom **123a1** in the vertical direction.

(65) Further, the tip of the upper supporting portion **131a** is formed in a hemispherical shape **131a1** which gradually becomes thinner upward (see FIG. 5B). The hemispherical shape **131a1** has the substantially same curvature as a concave surface of a curved bottom **123a1'** of the groove **123a** formed in the inner edge ring **123**. Therefore, the tip of the upper supporting portion **131a** is configured to be in line-contact with the groove **123a** formed in the inner edge ring **123**.

(66) The lower supporting portion **131b** at least has a diameter (the sixth width **W6** of FIG. 3) larger than the second width **W2** of the through hole **121a** formed in the insulating ring **121**. That is, the lower supporting portion **131b** has a step which protrudes outward in the radial direction from the outer periphery of the upper supporting portion **131a**, on the top surface thereof. In addition, the lower supporting portion **131b** is configured to support the inclined surface **1211a** formed in the lower hole portion **121a2** of the through hole **121a**, and therefore, supports the bottom of the insulating ring **121** and moves (lifts up) the bottom in the vertical direction. Further, in this case, the outer edge ring **122** held on the insulating ring **121** is also moved (lifted up) in the vertical direction at the same time.

(67) The actuator **132** moves the lift pin **131** in the vertical direction (axial direction) to lift the ring assembly **120** (insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123**) above the annular region **110b** (ring supporting surface). Therefore, the ring assembly **120** is transferred between the substrate support **11** and the transfer arm **2a** of the transfer apparatus **2**. One example of the actuator includes an electrical actuator or an air cylinder, a motor, etc.

(68) Further, the number of actuators **132** placed in the lifter **130** is not particularly limited. That is, for example, the plurality of lift pins **131** may be integrally moved in the vertical direction by one actuator **132**. Further, for example, a plurality of actuators **132** may be placed to correspond to the lift pins **131**, respectively, and the respective lift pins **131** may be independently moved in the vertical direction.

(69) Further, the substrate support **11** may include a temperature control module configured to control a temperature of at least one of the electrostatic chuck **112**, the ring assembly **120**, and the substrate **W** to a target temperature. The temperature control module may include a heater, a heat transfer medium, a path **111a**, or a combination thereof. In the flow path **111a**, a heat transfer fluid such as brine or gas flows. In an exemplary embodiment, the flow path **111a** is formed in the base

111 and one or more heaters are disposed in the ceramic member **112a** of the electrostatic chuck **112**. Further, the substrate support **11** may include a heat transfer gas supply configured to supply heat transfer gas (backside gas) to a gap between a back surface of the substrate **W** and the upper surface of the electrostatic chuck **112**.

(70) FIG. 2 is described again.

(71) The showerhead **13** is configured to introduce at least one processing gas from the gas supply **20** into the plasma processing space **10s**. The showerhead **13** includes at least one gas inlet **13a**, at least one gas diffusion space **13b**, and a plurality of gas introduction ports **13c**. The processing gas supplied from the gas supply **20** to the gas inlet **13a** is introduced into the plasma processing space **10s** from the plurality of gas introduction ports **13c** by passing through the gas diffusion space **13b**. Further, the showerhead **13** includes at least one upper electrode. Further, the gas introduction unit may include one or more side gas injectors (SGI) mounted on one or more openings formed on the side wall **10a** in addition to the showerhead **13**.

(72) The gas supply **20** may include at least one gas source **21** and at least one flow controllers **22**. In an exemplary embodiment, the gas supply **20** is configured to supply at least one processing gas to the showerhead **13** from the gas sources **21** corresponding to the one or more processing gas, respectively through the flow controllers **22** corresponding thereto, respectively. Each flow controller **22** may include, for example, a mass flow controller or a pressure control type flow controller. Further, the gas supply **20** may include at least one flow modulation device which modulates or pulses the flow of at least one processing gas.

(73) The power supply **30** includes an RF power supply **31** coupled to the plasma processing chamber **10** through at least one impedance matching circuit. The RF power supply **31** is configured to supply at least one RF signal (RF power) to at least one lower electrode and/or at least one upper electrode. Therefore, plasma is generated from at least one processing gas supplied to the plasma processing space **10s**. Therefore, the RF power supply **31** may serve as at least a part of the plasma generator **12**. Further, by supplying the bias RF signal to at least one lower electrode, a bias potential may be generated on the substrate **W** and ion components in the formed plasma may be attracted to the substrate **W**.

(74) In an exemplary embodiment, the RF power supply **31** includes a first RF generator **31a** and a second RF generator **31b**. The first RF generator **31a** is configured to be coupled to at least one lower electrode and/or at least one upper electrode through at least one impedance matching circuit, and to generate a source RF signal (source RF power) for plasma generation. In an exemplary embodiment, the source RF signal has a frequency in a range of 10 MHz to 150 MHz. In an exemplary embodiment, the first RF generator **31a** may be configured to generate a plurality of RF signals having different frequencies. One or more generated source RF signals are supplied to at least one lower electrode and/or at least one upper electrode.

(75) The second RF generator **31b** is configured to be coupled to at least one lower electrode through at least one impedance matching circuit, and to generate the bias RF signal (bias RF power). The frequency of the bias RF signal may be the same as or different from the frequency of the source RF signal. In an exemplary embodiment, the bias RF signal has a frequency lower than the source RF signal. In an exemplary embodiment, the source RF signal has a frequency in a range of 100 kHz to 60 MHz. In an exemplary embodiment, the second RF generator **31b** may be configured to generate a plurality of bias RF signals having different frequencies. One or more generated bias RF signals are supplied to at least one lower electrode. Further, in various exemplary embodiments, at least one of the source RF signal and the bias RF signal may be pulsed.

(76) Further, the power supply **30** may include the DC power supply **32** coupled to the plasma processing chamber **10**. The DC power supply **32** includes a first DC generator **32a** and a second DC generator **32b**. In an exemplary embodiment, the first DC generator **32a** is configured to be connected to at least one lower electrode and generate a first DC signal. The generated first DC signal is applied to at least one lower electrode. In an exemplary embodiment, the second DC

generator **32b** is configured to be connected to at least one upper electrode and generate a second DC signal. The generated second DC signal is applied to at least one upper electrode.

(77) In various exemplary embodiments, the first and second DC signals may be pulsed. In this case, a sequence of voltage pulses is applied to at least one lower electrode and/or at least one upper electrode. The voltage pulse may have a pulse waveform of a rectangle, a trapezoid, a triangle, or a combination thereof. In an exemplary embodiment, a waveform generator for generating a sequence of voltage pulses from the DC signal is connected between the first DC generator **32a** and at least one lower electrode. Therefore, the first DC generator **32a** and the waveform generator constitute a voltage pulse generator. When the second DC generator **32b** and the waveform generator constitute the voltage pulse generator, the voltage pulse generator is connected to at least one upper electrode. The voltage pulse may have a positive polarity and have a negative polarity. Further, the sequence of voltage pulses may include one or more positive voltage pulses and one or more negative voltage pulses within one cycle. Further, the first and second DC generators **32a** and **32b** may be provided in addition to the RF power supply **31**, or the first DC generator **32a** may be provided instead of the second RF generator **31b**.

(78) The exhaust system **40** may be connected to a gas outlet **10e** provided on a bottom of the plasma processing chamber **10**, for example. The exhaust system **40** may include a pressure adjustment valve and a vacuum pump. By the pressure adjustment valve, a pressure in the plasma processing space **10s** is adjusted. The vacuum pump may include a turbo molecule pump, a dry pump, or a combination thereof.

(79) Hereinabove, various exemplary embodiments have been described, but the present disclosure is not limited to the exemplary embodiment, but various additions, omissions, substitutions, and modifications may be made. Further, it is possible to form another exemplary embodiment by combining elements in different exemplary embodiments.

(80) For example, in the exemplary embodiment, the tip of the upper supporting portion is formed in the hemispherical shape **131a1** as illustrated in FIG. 3, but the tip of the upper supporting portion **131a** may substantially have a flat surface **131a2** as illustrated in FIG. 6A. In this case, in order to contact the tip of the upper supporting portion **131a** and the bottom of the groove **123a**, the bottom shape of the groove **123a** also preferably has a flat bottom **123a1''** which engages with the tip of the upper supporting portion **131a** instead of the curved bottom **123a1'**.

(81) Further, in this case, as illustrated in FIG. 6B, at least three grooves **123a** of the conductive ring **123** may be defined by the flat bottom **123a1''**, a vertical side surface **123a3** which extends downward from the flat bottom **123a1''**, and a flare-shaped side surface **123a2** which spreads downward from the vertical side surface **123a3**.

(82) Further, for example, in the exemplary embodiment, the through hole **121a** formed in the insulating ring **121** is formed in the substantially rectangular shape, and preferably formed in the substantially square shape, but as illustrated in FIG. 6C, the through hole **121a** may be formed in the substantially long hole shape in which a round shape of four corners is large. In this case, the through hole **121a** having the substantially long hole shape preferably has the long hole shape which has the first width **W1** which extends in the radial direction **R1** of the insulating ring **121** is larger than the second width **W2** which extends in the circumferential direction **C1**, i.e., has a long-side direction as the radial direction **R1** of the insulating ring **121**.

(83) <Effects, Etc.>

(84) The plasma processing system according to an exemplary embodiment is configured as above. That is, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber **10**, a substrate support **11**, an insulating ring **121**, a conductive ring **123**, at least three lift pins **131**, and at least one actuator **132**. The substrate support **11** is disposed in the plasma processing chamber **10**, and has a substrate supporting surface **110a** and a ring supporting surface **110b**. The insulating ring **121** is placed on the ring supporting surface **110b**, and has at least three through holes **121a**. Each of at least three through holes **121a** has an upper hole portion **121a1** and

a lower hole portion **121a2**. The upper hole portion **121a1** has the first width **W1** in the radial direction **R1** of the insulating ring **121** and the second width **W2** in the circumferential direction **C1** of the insulating ring **121**. The second width **W2** is smaller than the first width **W1**, and the lower hole portion **121a2** has a flare shape which spreads downward when viewed from the side. The conductive ring **123** is supported by the insulating ring **121**, and has at least three grooves **123a** corresponding to at least three through holes **121a**, respectively on the bottom thereof. Each of at least three grooves **123a** has the third width **W3** in the radial direction **R2** of the conductive ring **123** and the fourth width **W4** in the circumferential direction **C2** of the conductive ring **123**. The fourth width **W4** is smaller than the third width **W3**. At least three lift pins **131** are placed below the ring supporting surface **110b**, and correspond to at least three grooves **123a**, respectively. Each of at least three lift pins **131** has the upper supporting portion **131a** and the lower supporting portion **131b**. The upper supporting portion **131a** is configured to support the conductive ring **123** from downward by contacting the bottom **123a1** of the groove **123a** of the conductive ring **123** via the through hole **121a** of the insulating ring **121**. The lower supporting portion **131b** is configured to support the insulating ring **121** from downward by contacting the inclined surface **1211a** of the insulating ring **121** defining the lower hole portion **121a2**. At least one actuator **132** is configured to move at least three lift pins **131** in the vertical direction. The lower supporting portion **131b** of the lift pin **131** is configured to 2-point-contact the inclined surface **1211a** in the radial direction **R1** of the insulating ring **121** without contacting the inclined surface **1211a** in the circumferential direction **C1** of the insulating ring **121**.

(85) In an exemplary embodiment, each of at least three through holes **121a** of the insulating ring **121** has the rectangular shape when viewed from upward. In an exemplary embodiment, the upper supporting portion **131a** has a cylindrical shape having a first diameter **W5** smaller than the second width **W2**, and the lower supporting portion **131b** has a cylindrical shape having a second diameter **W6** larger than the second width **W2**. In an exemplary embodiment, the tip of the upper supporting portion **131a** of the lift pin **131** has the hemispherical shape **131a1**. In an exemplary embodiment, the tip of the upper supporting portion **131a** of the lift pin **131** has the flat surface **131a2**. In an exemplary embodiment, each of at least three grooves **123a** of the conductive ring **123** is defined by the flat bottom **123a1''**, the vertical side surface **123a3** which extends downward from the flat bottom **123a1''**, and the flare-shaped side surface **123a2** which spreads downward from the vertical side surface **123a3**. In an exemplary embodiment, each of at least three grooves **123a** of the conductive ring **123** is defined by the curved bottom **123a1'**. In an exemplary embodiment, each of at least three grooves **123a** of the conductive ring **123** is defined by the flat bottom **123a1''** and the flare-shaped side surface **123a2** which spreads downward from the flat bottom **123a1''**. In an exemplary embodiment, the plasma processing system **1** further includes an additional conductive ring **122** placed to surround around the conductive ring **123** and the insulating ring **121**. Further, at least a part of the conductive ring **122** overlaps with an external portion of the insulating ring **121** in the vertical direction.

(86) Further, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber **10**, a substrate support **11**, a first annular member **121**, a second annular member **123**, at least three lift pins **131**, and at least one actuator **132**. The substrate support **11** is placed in the plasma processing chamber **10**, and has a substrate supporting surface **110a** and a ring supporting surface **110b**. The first annular member **121** is placed on the ring supporting surface **110b**, and has at least three through holes **121a**. Each of at least three through holes **121a** has an upper hole portion **121a1** and a lower hole portion **121a2**. The upper hole portion **121a1** has a first width **W1** in a first direction **R1** and a second width **W2** in a second direction **C1** orthogonal to the first direction **R1**. The second width **W2** is smaller than the first width **W1**. The lower hole portion **121a2** has the flare shape **1211a** which spreads downward when viewed from the side. The second annular member **123** is supported by the first annular member **121**, and has at least three grooves **123a** corresponding to at least three through holes **121a**, respectively on the bottom

thereof. Each of at least three grooves **123a** has a third width **W3** in a third direction **R2** and a fourth width **W4** in a fourth direction **C2** orthogonal to the third direction **R2**. The fourth width **W4** is smaller than the third width **W3**. At least three lift pins **131** are placed below the ring supporting surface **110b**, and correspond to at least three grooves **123a**, respectively. Each of at least three lift pins **131** has the upper supporting portion **131a** and the lower supporting portion **131b**. The upper supporting portion **131a** is configured to support the second annular member **123** from downward by contacting the bottom **123a1** of the groove **123a** of the second annular member **123** via the through hole **121a** of the first annular member **121**. The lower supporting portion **131b** is configured to support the first annular member **121** from downward by contacting the inclined surface **121a1** of the first annular member **121** defining the lower hole portion **121a2**. At least one actuator **132** is configured to move at least three lift pins **131** in the vertical direction.

(87) In the plasma processing system, the substrate **W** is loaded into the plasma processing apparatus **1** by using the transfer apparatus **2**, and desired plasma processing such as etching is performed for the loaded substrate **W**, and then the substrate **W** is unloaded to the outside of the plasma processing apparatus **1** by using the transfer apparatus **2**.

(88) In the exemplary embodiment, the inner edge ring **123** and the outer edge ring are made of a conductive material. Therefore, at the time of the plasma processing in the plasma processing system, a sheath is formed above the ring assembly **120**, more specifically, above the inner edge ring **123** as the conductive ring and the outer edge ring **122** as the additional conductive ring. Therefore, an excellent and uniform plasma processing result may be obtained at the center and the periphery of the substrate **W** held on the substrate support **11**.

(89) Here, as described above, in recent years, the plasma processing system, enhancement of installation precision of the ring assembly on the substrate support **11** is required.

(90) In the Japanese Laid-open Patent Publication, it is disclosed that in order to enhance the installation precision of the annular member, the groove formed in the edge ring and the through hole formed in the cover ring are formed to be larger than the transfer precision of the annular member, but in this case, for example, there is a risk that the positioning precision of the annular member will deteriorate when a relative dislocation occurs at positions of the lifter and the groove (through hole) due to a mechanical difference (e.g., processing tolerance) or thermal expansion of the annular member.

(91) In this point, by the plasma processing apparatus **1** according to the technique of the present disclosure, the through hole **121a** is formed in the insulating ring **121** corresponding to the cover ring in a substantially rectangular shape in which the first width **W1** is larger than the second width **W2**, and the groove **123a** is also formed in the inner edge ring **123** in a substantially long hole shape in which the third width **W3** is larger than the fourth width **W4**.

(92) Therefore, even though the position of the lift pin **131** and the position of the through hole **121a** (groove **123a**) are dislocated due to the processing tolerance, the insulating ring **121** is slidably movable with the lift pin **131** as an axis, i.e., becomes a free state in the long-side direction of the through-hole **121a** (groove **123a**), so the insulating ring **121** may be appropriately guided to a desired location. Further, in this case, since the outer edge ring **122** as the additional conductive ring is held on the upper surface of the insulating ring **121**, the location of the outer edge ring **122** is simultaneously corrected.

(93) Further, in this case, as described above, the lower supporting portion **131b** of the lift pin **131** contacts two points on the inclined portion of the lower hole portion **121a2** of the through hole **121a**. As a result, as compared with a case where the lower supporting portion **131b** and the inclined portion of the lower hole portion **121a2** are in contact with each other at three points or more, or in line-contact with each other, and in surface-contact with each other, the insulating ring **121** easily slidably moves to more appropriately execute dislocation correction of the insulating ring **121** (ring assembly **120**).

(94) Further, in this case, as described above, the upper supporting portion **131a** of the lift pin **131**

is in line-contact with the bottom **123a1** of the groove **123a**.

(95) When the upper supporting portion **131a** and the bottom **123a1** of the groove **123a** are in contact with each other only at one point or two points, since the upper supporting portion **131a** and the bottom **123a1** slidably move relatively easily, the dislocation correction of the inner edge ring **123** may be facilitated, but meanwhile, since the upper supporting portion **131a** and the bottom **123a1** slidably move easily, the dislocation of the inner edge ring **123** easily occurs when the lift pin **131** moves in the vertical direction.

(96) Meanwhile, when the upper supporting portion **131a** and the bottom **123a1** of the groove **123a** are in surface-contact with each other, relative slidable movement of the bottom **123a1** and the upper supporting portion **131a** may be suppressed, but meanwhile, since the slidable movement is suppressed, it is difficult to correct the dislocation of the inner edge ring **123**.

(97) In this point, in the above exemplary embodiment, since the upper supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** of the groove **123a**, the bottom **123a1** slidably moves on the upper supporting portion **131a** to appropriately execute the dislocation correction of the inner edge ring **123** and at the same time, the dislocation of the inner edge ring **123** may also be appropriately suppressed when the lift pin **131** moves in the vertical direction.

(98) Further, when the through hole **121a** is formed in the insulating ring **121**, the through hole having the substantially rectangular shape in which the first width **W1** is larger than the second width **W2** as such, the insulating ring **121** slidably moves in the long-side direction of the through hole **121a** as described above, and as a result, there is a risk that the location of the insulating ring **121** will be dislocated when the lift pin **131** moves in the vertical direction.

(99) By considering such a point, the through hole **121a** is preferably formed in the substantially square shape as described above in order to suppress the dislocation of the insulating ring **121** (enhance the placement precision of the insulating ring **121**). Further, in this case, the long side-direction width (second width **W2**) of the through hole **121a** is preferably set to a size in which the lower supporting portion **131b** of the lift pin **131** contacts the inclined portion of the through hole **121a** at two points.

(100) As a result, the lower supporting portion **131b** and the inclined portion contact at least two points each other to execute the dislocation correction of the insulating ring **121**, and at the same time, a free width of the lift pin **131** in the long-side direction of the through hole **121a** decreases to suppress the dislocation of the insulating ring **121** when the lift pin **131** moves in the vertical direction.

(101) <Replacement Method of Ring Assembly>

(102) Next, replacement processing of the ring assembly **120** configured as above will be described by using drawings. Further, the following processing is executed under the control by the controller **3** as an example.

(103) (Step St1: Delivery and Unloading of Inner Edge Ring **123**)

(104) First, a consumed inner edge ring **123** to be replaced is delivered to the transfer apparatus **2** from the substrate support **11**.

(105) Specifically, first, ascending of the lifter **130** is executed, and as illustrated in FIG. 7, the inner edge ring **123** is delivered to the upper supporting portion **131a** of the lifter **130** passing through the through hole **113a** of the support **113** and the through hole **121a** of the insulating ring **121** from the upper surface of the insulating ring **121**. In this case, the ascending of the lifter **130** is executed in a range in which the insulating ring **121** is not delivered by the lower supporting portion **131b**, and further, executed until the tip of the upper supporting portion **131a** reaches a desired height **H1**. Here, the desired height **H1** is a height of the upper supporting portion **131a** of the lifter **130** based on the upper surface of the outer edge ring **122** held on the insulating ring **121** mounted on the support **113**. Further, the desired height **H1** is a height at which the transfer arm **2a** does not interfere with the outer edge ring **122** and the inner edge ring **123** when the transfer arm **2a** is inserted and extracted between the outer edge ring **122** on the insulating ring **121** mounted on

the support **113** and the inner edge ring **123** supported on the upper supporting portion **131a**.

(106) Next, the inner edge ring **123** is unloaded from the plasma processing chamber **10** of the plasma processing apparatus **1**.

(107) Specifically, the transfer arm **2a** is inserted into the decompressed plasma processing chamber **10** via a loading/unloading port (not illustrated). In addition, the transfer arm **2a** moves between the outer edge ring **122** on the insulating ring **121** mounted on the support **113** and the inner edge ring **123** supported on the upper supporting portion **131a** of the lifter **130**.

(108) Subsequently, descending of the lifter **130** is executed, and the inner edge ring **123** is delivered to the transfer arm **2a** from the upper supporting portion **131a** of the lifter **130**. Next, the transfer arm **2a** is extracted from the plasma processing chamber **10**, and the inner edge ring **123** itself is unloaded to the outside of the plasma processing apparatus **1**. The unloaded inner edge ring **123** is loaded into a receiving module (not illustrated) as an example.

(109) (Step St2: Delivery and Unloading of Insulating Ring **121** and Outer Edge Ring **122**)

(110) When the inner edge ring **123** is unloaded from the plasma processing apparatus **1**, the consumed insulating ring **121** and outer edge ring **122** to be replaced are subsequently delivered to the transfer apparatus **2** from the substrate support **11**.

(111) Specifically, first, the ascending of the lifter **130** is executed, and as illustrated in FIG. **8**, the insulating ring **121** is delivered to the lower supporting portion **131b** of the lifter **130** passing through the through hole **113a** of the support **113** from the upper surface of the annular region **110b** (ring supporting surface) of the substrate support **11**. In this case, since the outer edge ring **122** is held on the insulating ring **121**, the outer edge ring **122** is also delivered by the lower supporting portion **131b** via the insulating ring **121**. In this case, the ascending of the lifter **130** is executed until the step of the lower supporting portion **131b** reaches a desired height H2. Here, the desired height H2 is a height of the lower supporting portion **131b** of the lifter **130** based on the upper surface (more specifically, the upper surface of the electrostatic chuck **112**) of the substrate support **11**. Further, the desired height H2 is a height at which the transfer arm **2a** does not interfere with the insulating ring **121** and the electrostatic chuck **112** when the transfer arm **2a** is inserted and extracted between the electrostatic chuck **112** and the insulating ring **121** supported on the lower supporting portion **131b**.

(112) Next, the insulating ring **121** and the outer edge ring **122** are unloaded from the plasma processing chamber **10** of the plasma processing apparatus **1**.

(113) Specifically, the transfer arm **2a** is inserted into the decompressed plasma processing chamber **10** via a loading/unloading port (not illustrated). In addition, the transfer arm **2a** moves between the electrostatic chuck **112**, and the insulating ring **121** supported on the lower supporting portion **131b** of the lifter **130**.

(114) Subsequently, the descending of the lifter **130** is executed, and the insulating ring **121** and the outer ring **122** are delivered to the transfer arm **2a** from the lower supporting portion **131b** of the lifter **130**. Next, the transfer arm **2a** is extracted from the plasma processing chamber **10**, and the insulating ring **121** and the outer edge ring **122** are unloaded to the outside of the plasma processing apparatus **1**. The unloaded insulating ring **121** and the outer edge ring **122** are loaded into a receiving module (not illustrated) as an example.

(115) (Step St3: Loading and Mounting of Ring Assembly **120**)

(116) When the inner edge ring **123**, the insulating ring **121**, and the outer edge ring **122** are unloaded from the plasma processing apparatus **1**, a new ring assembly **120** (the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123**) for replacement are subsequently loaded into the plasma processing apparatus **1**.

(117) Specifically, the transfer arm **2a** of the transfer apparatus **2**, which holds the new ring assembly **120** including the insulating ring **121**, and the outer edge ring **122** and the inner edge ring **123** held on the insulating ring **121** is inserted into the plasma processing chamber **10** via the loading/unloading port (not illustrated). In this case, the plasma processing chamber **10** may be

decompressed. In addition, as illustrated in FIG. 9, the new ring assembly **120** is transferred by the transfer arm **2a** above the annular region **110b** of the substrate support **11**.

(118) Next, the new ring assembly **120** is mounted on the annular region **110b** of the substrate support **11** from the transfer apparatus **2**.

(119) Specifically, the ascending of the lifter **130** is executed, and as illustrated in FIG. 10A, the inner edge ring **123** is delivered to the upper supporting portion **131a** of the lifter **130** passing through the through hole **121a** of the insulating ring **121** from the upper surface of the insulating ring **121** held on the transfer arm **2a**.

(120) In this case, in the through hole **121a** of the insulating ring **121** and the groove **123a** of the inner edge ring **123**, as described above, the lower end of the inclined portion is formed to be at least larger than the transfer precision of the inner edge ring **123** by the transfer apparatus **2**. As a result, even though the lift pin **131**, the through hole **121a**, and the groove **123a** are dislocated due to the transfer precision of the transfer apparatus **2**, the lift pin **131** is appropriately inserted into the through hole **121a** to be further guided to the bottom **123a1** of the groove **123a**.

(121) Subsequently, the ascending of the lifter **130** is continued, and the insulating ring **121** is delivered to the lower supporting portion **131b** of the lifter **130** from the transfer arm **2a** as illustrated in FIG. 10B. In this case, the outer edge ring **122** on the insulating ring **121** is also simultaneously delivered to the lower supporting portion **131b**. In this case, the ascending of the lifter **130** is executed until the step of the lower supporting portion **131b** reaches a desired height **H2**.

(122) In this case, in the exemplary embodiment, in the through hole **121a** of the insulating ring **121** and the groove **123a** of the inner edge ring **123**, a diameter-direction width (a first width **W1** or a third width **W3**) is larger a circumferential width (a second width **W2** or a fourth width **W4**). As a result, even though the lift pin **131**, the through hole **121a**, and the groove **123a** are dislocated due to the processing tolerance, the lift pin **131** is appropriately inserted into the through hole **121a** to be further guided to the bottom **123a1** of the groove **123a**.

(123) Further, in this case, the lower supporting portion **131b** of the lift pin **131** contacts the inclined portion of the through hole **121a** at two points, and as a result, the insulating ring **121** becomes a free state in the long-side direction of the through hole **121a** with the lift pin **131** as the axis. As a result, even though the lift pin **131**, the through hole **121a**, and the groove **123a** are dislocated due to various causes, the insulating ring **121** slidably moves in the long-side direction of the through hole **121a** to appropriately position the insulating ring **121** and the outer edge ring **122** at desired locations.

(124) Further, similarly, the upper supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** of the groove **123a**, and as a result, the inner edge ring **123** may be appropriately positioned at a desired location, and at the same time, the dislocation of the inner edge ring **123** at the time of subsequent mounting of the ring assembly **120** may be suppressed.

(125) Next, extraction of the transfer arm **2a** from the plasma processing chamber **10** is executed. Further, the descending of the lifter **130** is executed. As a result, the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123** are mounted on the upper surface of the annular region **110b** of the substrate support **11** and the upper surface of the support **113**. Specifically, first, the insulating ring **121** and the outer edge ring **122** are mounted on the upper surface of the support **113**, and next, the inner edge ring **123** is mounted on the upper surface of the insulating ring **121** and the upper surface of the annular region **110b** of the substrate support **11**.

(126) Therefore, a series of processing for mounting the ring assembly **120** on the substrate support **11** is completed.

(127) Further, in the exemplary embodiment, loading and mounting of the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123** as the ring assembly **120** are simultaneously executed, but loading and mounting of any one of the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123** may be independently executed.

(128) Further, in the exemplary embodiment, delivery and unloading of the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123** as the ring assembly **120** are sequentially executed, but delivery and unloading of any one of the insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123** may be simultaneously executed. In this case, in a state in which the inner edge ring **123** is mounted on the insulating ring **121**, i.e., the delivery between the lift pin **131** and the transfer arm **2a** and unloading from the plasma processing apparatus **1** may be executed integrally with the insulating ring **121**.

(129) <Modified Example of Edge Ring>

(130) FIGS. **11A** to **11C** are diagrams illustrating a configuration example of an inner edge ring **223** according to another exemplary embodiment.

(131) In the inner edge ring **223** according to another exemplary embodiment, among a plurality of, in the illustrated example, three grooves **223a** to **223c** formed on the bottom, at least one groove (a groove **223a** in the illustrated example) may be formed in a circular shape (FIG. **11B**) and the remaining grooves (grooves **223b** and **223c** in the illustrated example) may be formed in a long hole shape (FIG. **11C**).

(132) In an exemplary embodiment, the groove **223a** having the circular shape has the substantially same diameter (a first diameter **D1** of FIGS. **11A** and **11B**) as the width (fifth width **W5**) of the upper supporting portion **131a** of the lift pin **131** on a bottom **223a1**. Further, the bottom **223a1** of the groove **223a** has a shape of being fitted into a tip shape of the upper supporting portion **131a** of the lift pin **131**. In other words, the tip of the upper supporting portion **131a** is configured to be in surface-contact with the groove **223a1** of the groove **223a** formed in the inner edge ring **223**.

(133) Further, a side surface **223a2** of the groove **223a** has an inclined portion is so that the first diameter **D1** of the bottom **223a1** spreads downward (toward the bottom of the inner edge ring **223**). A lower end of the inclined portion of the groove **223a** is preferably formed to be at least larger than the transfer precision of the inner edge ring **223** by the transfer apparatus **2**.

(134) In an exemplary embodiment, the grooves **223b** and **223c** having the long hole shape have the same structure as the groove **123a** of the inner edge ring **123** illustrated as the exemplary embodiment. That is, the grooves **223b** and **223c** are formed in which a third width **W3** which extends in the radial direction of the inner edge ring **223** is larger than a fourth width **W4** which extends in the circumferential direction of the inner edge ring **223** orthogonal to the radial direction, and bottoms **223b1** and **223c1** have, in particular, a shape of being fitted into the tip shape of the upper supporting portion **131a** of the lift pin **131** in an extension direction of the fourth width **W4**. Further, the inclined portions are formed on side surfaces **223b2** and **223c2** of the grooves **223b** and **223c**. A lower end of the inclined portion of the groove **123a** is preferably formed to be at least larger than the transfer precision of the inner edge ring **123** by the transfer apparatus **2**.

(135) As illustrated as the exemplary embodiment, when all of three grooves **123a** formed on the bottom of the inner edge ring **123** have the long hole shape, the inner edge ring **123** becomes the free state in the long-side direction of the groove **123a** as described above, so there is a risk that the inner edge ring **123** slidably moves to be dislocated.

(136) Therefore, in the inner edge ring **223** according to another exemplary embodiment, at least one groove **223a** is formed in the circular shape, and positioning of the inner edge ring **23** is executed based on the circular groove **223a**. That is, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber **10**, a substrate support **11**, an annular member **223**, at least three lift pins **131**, and at least one actuator **132**. The substrate support **11** is placed in the plasma processing chamber **10**, and has a substrate supporting surface **110a** and a ring supporting surface **110b**. The annular member **223** is placed on the ring supporting surface **110b**, and has at least three grooves **223a** to **223c**. A first groove **223a** among at least three grooves **223a** to **223c** has the circular bottom **223a** when viewed from upward. A second groove **223b** and a third groove **223c** among at least three grooves **223a** to **223c** have the bottoms **223b1** and **223c1**

having a first width W3 in a radial direction R3 of the annular member 223 and a second width W4 in a circumferential direction C3 of the annular member 223. The second width W4 is smaller than the first width W3. At least three cylindrical lift pins 131 are placed below the ring supporting surface 110b, and correspond to at least three grooves 223a to 223c, respectively. At least three cylindrical lift pins 131 are configured to support the annular member 223 from downward by contacting the bottoms of the grooves 223a to 223c of the annular member 223, respectively. At least one actuator 132 is configured to move at least three cylindrical lift pins 131 in the vertical direction.

(137) Therefore, the tip of the upper supporting portion 131a and the bottom 223a1 of the groove 223a having the substantially same size are fitted to be in surface-contact with each other, so the inner edge ring 223 and the lifter 130 are at least suppressed from slidably moving in a horizontal direction. Further, since the remaining grooves 223b and 223c are formed in the long hole shape, even though the grooves 223b and 223c, and the lift pin 131 are dislocated due to various causes, the lift pin 131 may be appropriately guided to the grooves 223b and 223c by correcting the dislocation and the inner edge ring 223 may be positioned at a desired location.

(138) Further, in the exemplary embodiment, among at least three grooves 223a to 223c, one groove 223a is formed in the circular shape, and the remaining grooves 223b and 223c are formed in the long hole shape, but for example, when the positioning of the inner edge ring 223 may be appropriately executed, each of at least three grooves 223a to 223c may be formed in the circular shape.

(139) In other words, in an exemplary embodiment, each of at least three grooves 223a to 223c of the conductive ring 123 is defined by the bottom 223a1 having a shape to be fitted into the tip of the upper supporting portion 131a of the lift pin 131 and the flare-shaped side surface 223a2 which spreads downward from downward surface.

(140) <Modified Example of Groove Formed in Inner Edge Ring>

(141) Further, sectional shapes of long hole shaped grooves (the groove 123a of the inner edge ring 123 and the grooves 223b and 223c of the inner edge ring 223) formed in the inner edge ring are not limited to the exemplary embodiment.

(142) For example, as illustrated in FIG. 12A, the groove 123a may have the bottom 123a1", the vertical side surface 123a3 as a vertical wall which extends vertically downward from the flat bottom 123a1", and the flare-shaped side surface 123a2 as extended wall which spreads downward from the vertical side surface 123a3. In this case, a short side-direction fourth width W4 (a width of the vertical wide surface 123a3 part) of the groove 123a is preferably formed to be a little larger than the diameter (the fifth width W5) of the upper supporting portion 131a to slidably move with respect to the lift pin 131. In this case, a contact scheme of the upper supporting portion 131a and the groove 123a is a 1-point contact.

(143) For example, as illustrated in FIG. 12B, the groove 123a may be formed on a curved surface having a larger curvature than the tip (hemispherical shape 131a1) of the hemispherical lift pin 131. In other words, the groove 123a has only the curved bottom 123a1' having the curved shape, and may not include the vertical side surface 123a3 and the flare-shaped side surface 123a2. In this case, the groove 123a is configured to slidably move with respect to the lift pin 131. In this case, the contact scheme of the upper supporting portion 131a and the groove 123a is the 1-point contact.

(144) For example, as illustrated in FIG. 12C, the groove 123a may have the bottom 123a1" having the substantially flat surface and the flare-shaped side surface 123a2 as the extended wall which spreads downward from the flat bottom 123a1". In other words, as compared with the exemplary embodiment, in the example illustrated in FIG. 12C, the flat bottom 123a1" is not formed in a shape of being fitted into the lift pin 131. In this case, the contact scheme of the upper supporting portion 131a and the groove 123a is a 3-point contact.

(145) Further, in the examples illustrated in FIGS. 5 and FIGS. 12A to 12C above, the flare-shaped

side surface **123a2** which spread downward from downward surface **123a1** (or vertical side surface **123a3**) of the groove **123a** is configured by a side surface having a liner shape when viewed from the side, which spreads at a predetermined ratio downward, but the flare-shaped side surface **123a2** may be configured by the side surface having the curved shape when viewed from the side.

(146) Specifically, as illustrated in FIG. 12D as an example, the groove **123a** may have a curved bottom **123a1'** having the curved shape, which has a larger curvature than the tip (hemispherical shape **131a1**) of the hemispherical lift pin **131**, and a flare-shaped curved side surface **123a4** which spreads downward from the curved bottom **123a1'** and which is curved upward (toward an opposite side to the lift pin **131**). In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is the 1-point contact.

(147) Further, the curvature of the flare-shaped curved side surface **123a4** is not particularly limited. In other words, the groove **123a** is formed in the inner edge ring **123** provided in the plasma processing apparatus **1** according to the technique of the present disclosure, and the groove **123a** is constituted by the bottom **123a1**, and one or more side surfaces (side surfaces **123a2**, **123a3**, and **123a4**) having a predetermined curvature of "0" or more. The curvature of side surface being "0" refers to the side surface being the linear shape (side surface **123a2**) when viewed from the side. The curvature being larger than "0" refers to the side surface being the curved shape (side surface **123a4**) when viewed from the side.

(148) As such, the groove **123a** (grooves **223b** and **223c**) may be formed in a predetermined sectional shape. In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** varies depending on the sectional shape. From a viewpoint of facilitating the positioning of the upper supporting portion **131a** and the groove **123a**, a contact area of the upper supporting portion **131a** and the groove **123a** is preferably small. From a viewpoint of enhancing the installation precision of the upper supporting portion **131a** and the groove **123a**, the contact area of the upper supporting portion **131a** and the groove **123a** is preferably large.

(149) As such, the sectional shape of the groove **123a** (grooves **223b** and **223c**) may be appropriately designed according to a purpose.

(150) Further, the sectional shape of the circular groove (the groove **223a** of the inner edge ring **223**) formed in the inner edge ring is not also limited to the exemplary embodiment.

(151) For example, as illustrated in FIG. 13A, the groove **223a** may have a bottom **223a1''**, a vertical side surface **223a3** as a vertical wall which extends vertically downward from the flat bottom **223a1''**, and a flare-shaped side surface **223a2** as extended wall which spreads downward from the vertical side surface **223a3**. In this case, a first diameter D1 (a width of the vertical side surface (**223a3**) part) of the groove **223a** is preferably formed to be a little larger than the diameter (fifth width W5) of the upper supporting portion **131a** in order to appropriately contact the tip of the lift pin **131** with the flat bottom **223a1''**. Further, in this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is the 1-point contact.

(152) For example, as illustrated in FIG. 13B, the groove **223a** may be formed on a curved surface having a larger curvature than the tip of the hemispherical lift pin **131**. In other words, the groove **223a** has only the curved bottom **223a1'** having the curved shape, and may not include the flare-shaped side surface **223a2** and the vertical side surface **223a3**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **223a** is the 1-point contact.

(153) For example, as illustrated in FIG. 13C, the groove **223a** may have the bottom **223a1''** having the substantially flat surface and the flare-shaped side surface **223a2** as the extended wall which spreads downward from the flat bottom **223a1''**. In other words, as compared with the example illustrated in FIG. 11, in the example illustrated in FIG. 13C, the flat bottom **223a1''** is not formed in a shape of being fitted into the lift pin **131**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **223a** is the 3-point contact.

(154) Further, the illustrated is omitted, but even for the circular groove **223a**, similarly to the example illustrated in FIG. 12D, the flare-shaped side surface **123a2** may be configured by the

curve-shaped side surface when viewed from the side. In other words, the circular groove 223a may have a flare-shaped curved side surface not illustrated.

(155) Hereinabove, various exemplary embodiments have been described, but the present disclosure is not limited to the exemplary embodiment, but various additions, omissions, substitutions, and modifications may be made. Further, it is possible to form another exemplary embodiment by combining elements in different exemplary embodiments.

Claims

1. A plasma processing apparatus, comprising: a plasma processing chamber; a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface; an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a radial direction of the insulating ring and a second width in a circumferential direction of the insulating ring, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side; a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a radial direction of the conductive ring and a fourth width in a circumferential direction of the conductive ring, the fourth width being smaller than the third width; at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting portion being configured to support the conductive ring from downward by contacting a bottom of the groove of the conductive ring through the through hole of the insulating ring, the lower supporting portion being configured to support the insulating ring from downward by contacting an inclined surface of the insulating ring, the inclined surface defining the lower hole portion; and at least one actuator configured to vertically move the at least three lift pins.
2. The plasma processing apparatus of claim 1, wherein the lower supporting portion of the lift pin is configured to come into contact with two points on the inclined surface in the radial direction of the insulating ring without contacting the inclined surface in the circumferential direction of the insulating ring.
3. The plasma processing apparatus of claim 1, wherein each of the at least three through holes of the insulating ring has a rectangular shape when viewed from upward.
4. The plasma processing apparatus of claim 1, wherein the upper supporting portion has a cylindrical shape having a first diameter smaller than the second width, and the lower supporting portion has a cylindrical shape having a second diameter larger than the second width.
5. The plasma processing apparatus of claim 1, wherein a tip of the upper supporting portion of the lift pin has a hemispherical shape.
6. The plasma processing apparatus of claim 1, wherein a tip of the upper supporting portion of the lift pin has a flat surface.
7. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by: a flat bottom, a vertical side surface which extends downward from the flat bottom, and a flare-shaped side surface which spreads downward from the vertical side surface.
8. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by a curved bottom.
9. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by: a flat bottom, and a flare-shaped side surface which spreads

downward from the flat bottom.

10. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring has a bottom having a shape to be fitted into a tip of the upper supporting portion of the lift pin, and a flare-shaped side surface which spreads downward from the bottom.

11. The plasma processing apparatus of claim 1, further comprising: an additional conductive ring disposed to surround a circumference of the conductive ring and the insulating ring, wherein at least a part of the additional conductive ring overlaps an outer portion of the insulating ring in a vertical direction.

12. A plasma processing apparatus, comprising: a plasma processing chamber; a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface; an annular member disposed on the ring supporting surface, the annular member having at least three grooves on a lower surface, a first groove among the at least three grooves having a circular bottom when viewed from upward, a second groove and a third groove among the at least three grooves having a bottom having a first width in a radial direction of the annular member and a second width in a circumferential direction of the annular member, the second width being smaller than the first width; at least three cylindrical lift pins disposed below the ring supporting surface, the at least three cylindrical lift pins corresponding to the at least three grooves, respectively, each of the at least three cylindrical lift pins being configured to support the annular member from downward by contacting the bottom of the groove of the annular member; and at least one actuator configured to vertically move the at least three cylindrical lift pins.

13. The plasma processing apparatus of claim 12, wherein a tip of the cylindrical lift pin has a hemispherical shape.

14. The plasma processing apparatus of claim 12, wherein a tip of the cylindrical lift pin has a flat surface.

15. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by: a flat bottom, a vertical side surface which extends downward from the flat bottom, and a flare-shaped side surface which spreads downward from the vertical side surface.

16. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by a curved bottom.

17. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by: a flat bottom, and a flare-shaped side surface which spreads downward from the flat bottom.

18. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular conductive ring has a bottom having a shape to be fitted into a tip of an upper supporting portion of the cylindrical lift pin, and a flare-shaped side surface which spreads downward from the bottom.

19. A plasma processing apparatus, comprising: a plasma processing chamber; a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface; a first annular member disposed on the ring supporting surface, the first annular member having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a first direction and a second width in a second direction orthogonal to the first direction, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side; a second annular member supported by the first annular member, the second annular member having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a third direction and a fourth width in a fourth direction orthogonal to the third direction, the fourth width being smaller than the third

width; at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting portion being configured to support the second annular member from downward by contacting a bottom of the groove of the second annular member through the through hole of the first annular member, the lower supporting portion being configured to support the first annular member from downward by contacting an inclined surface of the first annular member, the inclined surface defining the lower hole portion; and at least one actuator configured to vertically move the at least three lift pins.
